

Current Status of the Simulation of Radiative Effects with McMule

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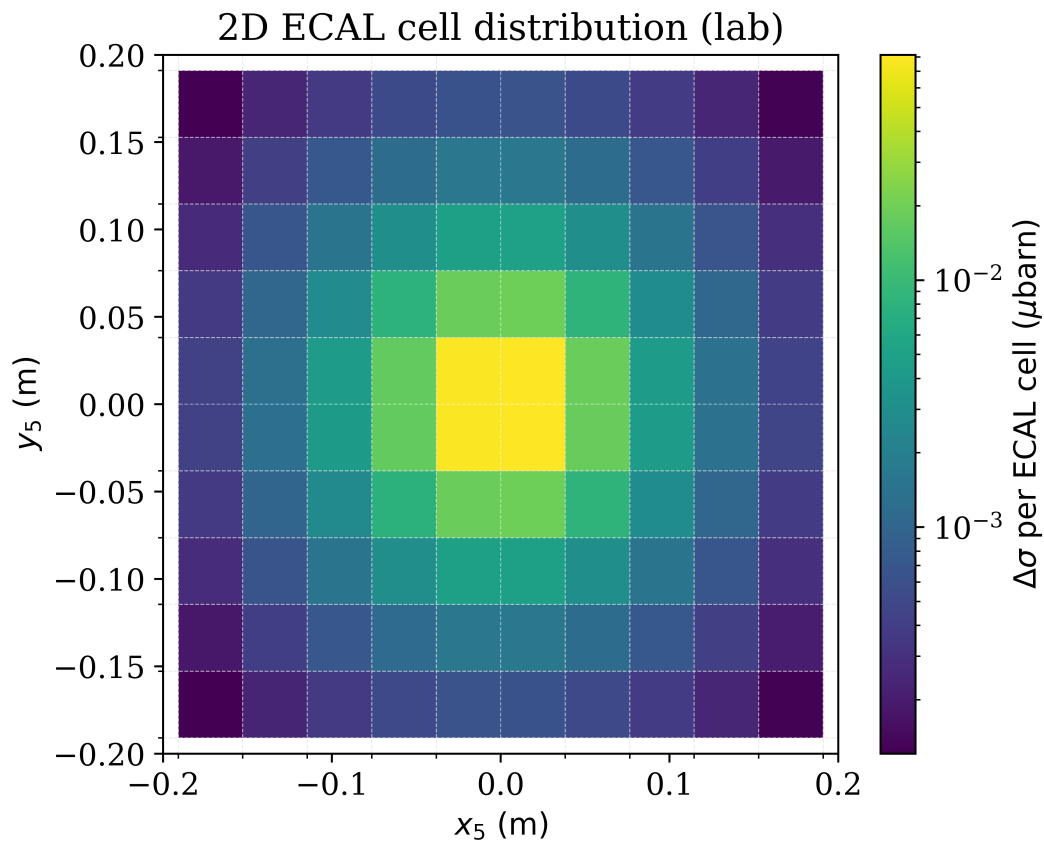


McMULE

Project Overview and Progress

- Topic: Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule
- Learning to work with McMule with support from Marco Rocco
- Simulation of LO ($\mu + p \rightarrow \mu + p$) and NLO processes
- Especially investigating the real-photon emission process
- Investigation of the impact of different cuts on the kinematic
- Cuts used to obtain the results in this presentation:
 - $0.3 < \theta_\mu < 2. [mrad]$
 - $E_\mu > 70 \text{ GeV}$
 - $-12. < \theta_\gamma < 12. [mrad]$

Expected Photon Cross Section



$0.3 < \theta_\mu < 2. [\text{mrad}]$
 $E_\mu > 70 \text{ GeV}$
 $-12. < \theta_\gamma < 12. [\text{mrad}]$

Expected Cross Sections and Count Rates

$$0.001 < Q^2 < 0.04 [GeV^2/c^2]$$

Photon Energy	100 MeV	200 MeV	500 MeV
σ_{LO} [mb]	0.2505664	0.2505670	0.2505621
R_{LO} [1/s]	31.902323	31.902393	31.901780
$\sigma_{R\gamma}$ [mb]	0.00011478	0.00010584	0.00008954
$R_{R\gamma}$ [1/s]	0.014614	0.013475	0.011401
σ_{NLO} [mb]	0.00286402	0.00285508	0.00283873
σ_{FULL} [mb]	0.25343043	0.25342204	0.25340088

Count rate input values:

Target Length: 60 cm

Hydrogen pressure: 20 bar

2.687×10^{19} Protons/cm³

Beam intensity: 2×10^6 1/s

$$0.3 < \theta_\mu < 2. [mrad]$$

$$E_\mu > 70 GeV$$

$$-12. < \theta_\gamma < 12. [mrad]$$

- Increasing the photon-energy threshold reduces both the real-photon cross section and count rate
- The real-photon process contributes at the $\sim 0.04\%$ level (depending on photon energy threshold)

Expected Cross Sections and Count Rates

$$0.001 < Q^2 < 0.04 [GeV^2/c^2]$$

Photon Energy	100 MeV	200 MeV	500 MeV
σ_{LO} [mb]	0.25055889	0.25056698	0.25056701
R_{LO} [1/s]	31.9013652	31.9023952	31.9023997
$\sigma_{R\gamma}$ [mb]	0.00011593	0.00009425	0.00010577
$R_{R\gamma}$ [1/s]	0.01476071	0.01200053	0.01346621
σ_{NLO} [mb]	0.00286519	0.00284407	0.00285465
σ_{FULL} [mb]	0.25342407	0.25341105	0.25342166

Count rate input values:

Target Length: 60 cm

Hydrogen pressure: 20 bar

2.687×10^{19} Protons/cm³

Beam intensity: 2×10^6 1/s

$$0.3 < \theta_\mu < 2. [mrad]$$

$$E_\mu > 70 GeV$$

$$-12. < \theta_\gamma < 12. [mrad]$$

BACKUP

Used Equations

- def calculate_rate(sigma_mb):
 ltarget = 60 #cm
 Hpres = 20 #bar
 Npvol = 2.687*1e19 #Protons/cm^3
 lbeam = 2*1e6 #1/s
 Np = Npvol * 2 * (Hpres/1.013) * ltarget #Protons/cm^2 (target thickness)

 rate = sigma_mb * 1e-27 * Np * lbeam
 return rate #1/s